Practical No. 3

Aim: Write a java program demonstrating classes and objects.

Code:

Program 1:

```
// Class Definition
class Student {
    // Instance variables (Attributes)
    String name;
    int rollNumber;
    double marks;

    // Constructor to initialize object state
    public Student(String name, int rollNumber, double marks) {
        this.name = name;
        this.rollNumber = rollNumber;
        this.marks = marks;
    }

    // Method to display student details
    public void displayDetails() {
        System.out.println("Roll No: " + rollNumber + " | Name: " + name + " | Marks: " + marks);
    }

    // Method to check pass/fail status
    public boolean hasPassed() {
        return marks >= 40.0;
    }
}

public class Main1 {
    public static void main(String[] args) {
        // Creating Object 1
        Student student1 = new Student("Alice", 101, 85.5);

        // Creating Object 2
        Student student2 = new Student("Bob", 102, 32.0);

        // Accessing object methods
        System.out.println("=== Student Records ===");
        student1.displayDetails();
        System.out.println("Passed: " + student1.hasPassed());

        System.out.println();

        student2.displayDetails();
        System.out.println("Passed: " + student2.hasPassed());
    }
}
```

Program 2:

```
import java.util.Scanner;
```

```
// Encapsulated Class
class BankAccount {
    // Private variables (Data Hiding)
    private String accountNumber;
    private double balance;


 // Getter for Balance
    public double getBalance() {
        return balance;
    }

    // Method to Deposit Money
    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Deposited $" + amount + ". New Balance: $" + balance);
        } else {
            System.out.println("Invalid deposit amount!");
        }
    }

    // Method to Withdraw Money
    public void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Withdrew $" + amount + ". Remaining Balance: $" + balance);
        } else {
            System.out.println("Withdrawal failed! Invalid amount or insufficient balance.");
        }
    }
}
```

Output:

Program1:

```
Initial Balance: $500.0
Deposited $150.0. New Balance: $650.0
Withdrew $200.0. Remaining Balance: $450.0
Withdrawal failed! Invalid amount or insufficient balance.
```

Program2:

```
=== Student Records ===
Roll No: 101 | Name: Alice | Marks: 85.5
Passed: true

Roll No: 102 | Name: Bob | Marks: 32.0
Passed: false
```